

Weibull Distribution

A continuous r.v. X ~~has~~ is said to have Weibull distribution with ~~two~~ three parameters ($c > 0$), $\alpha > 0$, β if the pdf is

$$f(x) = c\alpha^{-1} \left(\frac{x-\beta}{\alpha} \right)^{c-1} e^{-\left(\frac{x-\beta}{\alpha} \right)^c}$$

Appointments

The standard Weibull distribution is obtained when $\alpha = 1$, $\mu = 0$,
 $f(x) = cx^{c-1} e^{-x^c}$, $x > 0$, $c > 0$.

Phones to make

$$E(x^r) = \Gamma\left(\frac{r}{c} + 1\right)$$

$$\text{mean} = E(x) = \Gamma\left(\frac{1}{c} + 1\right)$$

$$\text{Var}(x) = \Gamma\left(\frac{2}{c} + 1\right) - \left\{ \Gamma\left(\frac{1}{c} + 1\right) \right\}^2$$

Remarks

c - Shape parameter

α - Scale parameter

β - Location parameter.

If the ~~loc~~ location parameter $\beta \rightarrow 0$
then ~~the~~ the distribution will be
two parameter Weibull distribution.

Now if

the shape parameter c is called
Weibull slope, because the value of
 c is equal to the slope of the line
in a probability plot.

If $c = 1$ then it reduces to
two parameter exponential.

c is a pure number.

Two parameter Weibull $\beta \rightarrow 0$

$$f(x) = \frac{c}{\alpha} \left(\frac{x}{\alpha}\right)^{c-1} e^{-\left(\frac{x}{\alpha}\right)^c}$$

$x \geq 0$

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When $c = 1$ then it belongs
exponential distribution

Appointments

mgf :

$$M_X(t) = E(e^{tx})$$

$$= \int_0^x e^{t \ln \frac{e}{2}} \left(\frac{x}{2}\right)^{e-t} \frac{e}{2} \left(\frac{x}{2}\right)^e dx$$

$$= \int_0^{\infty} e^{-tu} \frac{e^{-1} - 1}{e} u^e du$$

$$= \int_0^{\infty} \frac{x t u}{e^{x t u}} \frac{e^{-1-u}}{e^u} du = \int_0^{\infty} \frac{x t u e^{-1-u}}{e^{x t u} e^u} du$$

$$= \int_0^{\infty} \sum_{n=0}^{\infty} \frac{(x/t)^n}{n!} n e^{-x/t} dx$$

$$= \sum_{n=0}^{\infty} \frac{\alpha^n t^n}{n!} \Gamma\left(\frac{n}{c} + 1\right)$$

Phones to make

Remarks